Week 5 Feasibility Memo

1. Verdict

The dataset carries a genuine, near-complete triage label (ESI) and clinically sensible vital sign and chief-complaint patterns, but it is a single US health-system extract with an unexplained lack of missing data, and several caveats below must be resolved before any model output reaches a patient.

2. Dataset Summary

55,121 ED encounters, 225 features: demographics, vitals, arrival/disposition, 200 binary chief-complaint flags.

Target Label ESI

ESI	Encounters	Percentage
1 (Resuscitation)	77	0.1%
2 (Emergent)	19,924	32.5%
3 (Urgent)	27,010	49.0%
4 (Less Urgent)	8,896	16.1%
5 (Not Urgent)	1,214	2.2%

Most patients are in the mid-acuity range (ESI 2–3), which makes up 81% of the ESI. ESI 1 is rare, which is a constraint for modelling.

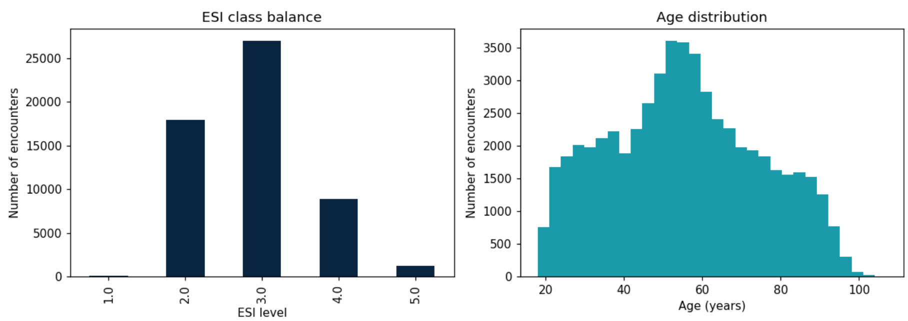

Figure 1. Left: ESI class balance the mid-acuity concentration and near-absence of ESI 1 are visible directly. Right: patient age distribution.

Temperature is recorded in Fahrenheit, not Celsius. This makes it easy for anyone to apply standard clinical thresholds.

3. Top 3 Concerns

- **A small number of vital signs readings were physically impossible.**

29 readings out of 55,121 visits well under 0.1%, mostly breathing rate and blood sugar were outside what a living patient could have, which could be data entry errors.

- **Two fields would let the model “cheat.”**

What happened to the patient afterward, and their prior disposition, are only known after the triage decision has already been made.

- **This is a single Academic hospital’s data**.

A Caribbean emergency department may see a different mix of patients, conditions and resources. Mitigation: treat any results as a benchmark only, local validation is required before this informs a Caribbean ED.

4. Top 3 Reasons to Proceed

- A real, non-synthetic triage label. Every patient in this dataset has an actual ESI score attached. The same 1 to 5 urgencies rating a triage nurse assign in real life. That means we have a genuine outcome to learn from and not a guess or stand in for one.

- Vitals move the way a clinician would expect as patients get sicker. As ESI level increases heart rate and breathing rate tend to go down and oxygen levels tend to go up, exactly the pattern closest to physiological studies. This means that the vital signs in this data are capturing real patient severity, not noise.

- The complaints people come in with match how urgent they are. Patients who show up with chest pain, trouble breathing or confusion tend to be marked more urgently. Patients who show up with a sore throat or a toothache tend to be marked less urgently. In other words, the data behaves the way a clinician would predict, which is a strong sign it is trustworthy enough to build.

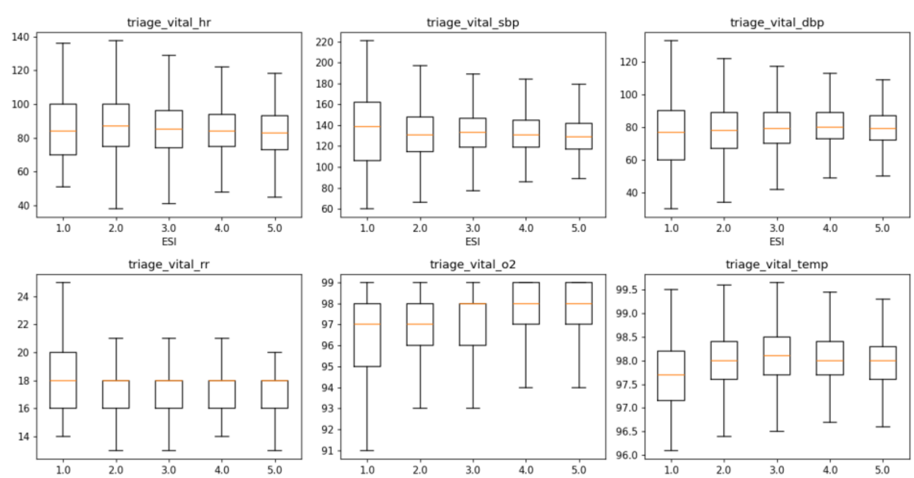

Figure 2. Vital signs are grouped by ESI level. Heart rate, respiratory rate, and oxygen saturation shift in the clinically expected direction as acuity increases, supporting the second reason above.

5. Caveats

- Temperature is recorded in Fahrenheit, not Celsius any threshold from a non-U.S. need converting first.

- The patient complaint flags are individually rare which means that they are best combined with the vital to give a proper ESI score.

- The patterns found so far (e.g. “Chest-pain visits tend to be more urgent”) are associations, not proof of cause and effect and not yet a working model.

- How well any of this holds up in an actual setting specifically in a Caribbean ED, has not been tested and should not be assumed.

Suggested Starting Features

1. Oxygen saturation - low oxygen is one of the fastest signs of a deteriorating patient.
2. Respiratory rate - breathing rate changes quickly with acuity.
3. Heart rate - a standard early-warning vital.
4. Systolic blood pressure - a standard early-warning vital.
5. Patient age - extremes of age - often track how urgent a visit turns out to be.
6. “Chest pain” flag - the complaint most associated with high urgency in this data.
7. “Shortness of breath” flag - second-most associated with high urgency.
8. “Suicidal ideation” flag - associated with high urgency.
9. “Alcohol intoxication” flag - associated with high urgency.
10. “Altered mental status” flag - associated with high urgency.

The first five are included on clinical grounds (established early-warning signs); the last five were ranked by how strongly they moved with urgency in this dataset. The two “what happened after triage” fields were deliberately left off this list, since they would not be available now a real triage decision is made.